CHAPTER 2

COLLECTION OF DATA

CHAPTER ESSENCE AT A GLANCE:
Master primary versus secondary data sources, survey instruments, questionnaire construction rules, and national statistical agencies (NSSO, Census of India).

4 CORE PRINCIPLES TO UNDERSTAND FIRST

- **Primary vs Secondary Data:** Primary data is first-hand information collected by an investigator for a specific inquiry; Secondary data has already been collected, processed, and published by other agencies.
- **Primary Data Collection Modes:** Methods include Direct Personal Interview, Indirect Oral Investigation, Information from Local Correspondents, Mailed Questionnaires, and Schedules sent through Enumerators.
- **Questionnaire Architecture Rules:** A good questionnaire contains limited questions, logical ordering from simple to complex, clear non-ambiguous wording, avoids leading or sensitive inquiries, and includes cross-checks.
- **Secondary Sources and Validation:** Published sources include government reports, international bodies (UN, IMF), and trade journals; secondary data must always be tested for reliability, suitability, and adequacy before use.

ESSENTIAL STATISTICAL FORMULAS & MEASURES

$$\text{Data Origin} \rightarrow \{\text{Primary: First-Hand Original}, \quad \text{Secondary: Existing Published}\}$$

Source-Based Information Taxonomy

$$\text{Survey Methods} = \text{Personal Interview} \longleftrightarrow \text{Mailed Questionnaire} \longleftrightarrow \text{Enumerator Schedule}$$

Spectrum of Primary Data Collection Instruments

$$\text{Pilot Survey} \Longleftrightarrow \text{Pre-Testing Questionnaire on Small Sample} \rightarrow \text{Eliminates Ambiguities}$$

Quality Control Protocol in Survey Design

$$\text{Secondary Data Scrutiny} \Longleftrightarrow \text{Reliability} \cap \text{Suitability} \cap \text{Adequacy}$$

Three Mandatory Triangulation Criteria for Published Data

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

- **Alert:** Never use secondary data without evaluating the collecting agency's credentials, scope, time reference, and definitions used.
- **Alert:** In mailed questionnaires, non-response bias is a critical hazard; always conduct a pilot survey and send polite follow-up reminders.
- **Alert:** Schedules filled by enumerators yield higher response rates and accuracy than self-administered mailed questionnaires, but cost substantially more.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Distinguish between Primary Data and Secondary Data on the basis of: (a) Originality, (b) Cost and Time, and (c) Need for verification. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **(a) Originality:** Primary data is original in nature as it is collected directly from respondents for the first time; Secondary data lacks originality as it has already been compiled by others. [1 Mark]

• **(b) Cost and Time:** Primary data collection requires significant financial expenditure, large field staff, and extensive time; Secondary data collection is economical and rapid. [1 Mark]

• **(c) Need for Verification:** Primary data fits the specific survey objective directly; Secondary data requires rigorous scrutiny for reliability, suitability, and adequacy. [1 Mark]

2. Explain any four essential qualities of a good questionnaire. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **1. Limited Number of Questions:** The questionnaire should be brief and concise to prevent respondent fatigue and maintain high completion rates. [1 Mark]

• **2. Clear and Simple Language:** Questions must be unambiguous, avoiding complex technical jargon and mathematically demanding questions. [1 Mark]

• **3. Logical Arrangement:** Questions should proceed logically from general introductory queries to specific detailed topics. [1 Mark]

• **4. Avoid Leading or Personal Questions:** Questions must not suggest answers (leading) or offend personal sensibilities regarding private wealth or habits. [1 Mark]

3. Compare Mailed Questionnaire Method and Schedules Sent Through Enumerators. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Mailed Questionnaire:** Sent by post or email with a covering letter and self-addressed envelope; completed independently by the respondent. Economical over wide geographic areas, but suffers from low response rates and misinterpretation. [1.5 Marks]

• **Schedules Sent Through Enumerators:** Enumerators visit respondents in person, interpret questions, and record answers directly. Yields high response rates and accurate data across illiterate populations, but is expensive and vulnerable to enumerator bias. [1.5 Marks]

4. What is a Pilot Survey? State its significance in statistical investigation. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Definition:** A preliminary survey conducted on a small sample before embarking on the large-scale main inquiry. [1 Mark]

• **Significance:** Pre-tests the questionnaire, uncovers ambiguous questions, assesses field costs and time duration, and trains enumerators for smooth survey execution. [1 Mark]

5. Name two premier statistical agencies in India and describe the primary function of each. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **National Sample Survey Office (NSSO):** Conducts nationwide socio-economic sample surveys on employment, consumer expenditure, industrial manufacturing, and housing conditions. [1.5 Marks]

• **Office of the Registrar General of India:** Conducts the decennial Population Census of India, recording complete demographic, occupational, literacy, and vital health statistics. [1.5 Marks]

6. What precautions must an investigator take before using Secondary Data? [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Precautions:** Verify the reliability of the collecting agency, assess whether the scope and objective of the original inquiry match current research, and ensure data definitions and units remain consistent. [2 Marks]